

KNOWLEDGE CHECK

Databricks Fundamental

Course quiz with answer key

The quiz has 40 multiple-choice questions, grouped by module. Choose the single best answer to each question. The answer key is on the last page.

MODULE 01

Platform and workspace

1. What are the three main layers of the Databricks lakehouse architecture?
 - A. Storage (Delta Lake), compute (clusters and SQL warehouses) and governance (Unity Catalog)
 - B. Bronze, silver and gold
 - C. Driver, executors and shuffle
 - D. HDFS, MapReduce and YARN
2. What is a SQL warehouse used for?
 - A. Compute optimised for SQL queries and BI tools
 - B. A long-running all-purpose cluster for notebooks
 - C. A storage container for Delta tables
 - D. A Unity Catalog object for managing permissions
3. Which magic command runs SQL in a Python notebook?
 - A. `%python`
 - B. `%sql`
 - C. `%run`
 - D. `%sh`
4. What is the key difference between DBFS and Unity Catalog volumes?
 - A. DBFS is faster for large files
 - B. Volumes are governed by Unity Catalog, with access control and lineage; DBFS is a deprecated, ungoverned file system that new workspaces no longer provide
 - C. DBFS supports the Delta format, while volumes support only Parquet
 - D. There is no difference
5. Which command creates a drop-down widget?
 - A. `dbutils.widgets.text()`
 - B. `dbutils.widgets.dropdown()`
 - C. `dbutils.widgets.choice()`
 - D. `dbutils.widgets.select()`

MODULE 02

Spark architecture

6. What is the driver responsible for in a Spark cluster?
 - A. Storing data on disk as the primary storage node
 - B. Running individual tasks in parallel
 - C. Coordinating execution: running your code, building the DAG and scheduling tasks
 - D. Compressing data before it is sent to the executors
7. What distinguishes a transformation from an action?
 - A. Transformations write data; actions read data
 - B. Transformations are lazy and return a new DataFrame; actions trigger execution
 - C. Transformations exist only in SQL; actions only in Python
 - D. Transformations run immediately; actions are lazy
8. What does adaptive query execution (AQE) do?
 - A. It is a manual tuning process carried out by an administrator
 - B. It loads data into memory before a query starts
 - C. It adjusts the execution plan at run time using actual statistics
 - D. It converts PySpark code to SQL
9. When does a shuffle occur?
 - A. Whenever a new DataFrame is created
 - B. Only when writing to Delta Lake
 - C. When data must be redistributed across partitions, typically in wide transformations such as `groupBy` or `join`
 - D. When `inferSchema` is used on CSV files
10. What does lazy evaluation mean?
 - A. Spark waits for free disk space before reading files
 - B. Spark does not execute transformations until an action is called
 - C. Spark caches every DataFrame after its first read
 - D. Spark uses a slower but more accurate engine

MODULE 03

ELT and ingestion

11. What is the main drawback of `inferSchema=True` on a large CSV file?
- A. The inferred schema is always wrong
 - B. Spark reads the file twice, once to infer the types and once to load the data
 - C. Only string columns are supported
 - D. It does not work with Unity Catalog
12. What does `explode()` do?
- A. Splits a string column on a delimiter
 - B. Raises an error on null values
 - C. Turns each element of an array column into a separate row
 - D. Flattens a struct into separate columns
13. Which join returns all rows from the left table and only the matching rows from the right?
- A. `INNER JOIN`
 - B. `LEFT JOIN`
 - C. `RIGHT JOIN`
 - D. `FULL OUTER JOIN`
14. How do you create a temporary view from a DataFrame?
- A. `df.saveAsTable("view_name")`
 - B. `df.createOrReplaceTempView("view_name")`
 - C. `spark.sql("CREATE VIEW view_name")`
 - D. `df.registerTable("view_name")`
15. What is the `abfss://` scheme used for in Azure Databricks?
- A. Reading public Amazon S3 buckets
 - B. Connecting to Azure SQL Database
 - C. Accessing Azure Data Lake Storage Gen2 through the Azure Blob File System driver
 - D. Downloading packages from PyPI
16. Which approach adds or updates a column using conditional logic?
- A. `df.filter()`
 - B. `df.withColumn()` with `when()` and `otherwise()`
 - C. `df.groupBy()`
 - D. `df.dropDuplicates()`

MODULE 04

Delta Lake

17. What does ACID stand for?

- A. Accurate, consistent, integrated, durable
- B. Atomicity, consistency, isolation, durability
- C. Automated, concurrent, incremental, distributed
- D. Analytical, columnar, indexed, distributed

18. What is the Delta transaction log (`_delta_log`)?

- A. A Spark UI page with query history
- B. A directory of CSV files with row-level changes
- C. A JSON log that records every change to a Delta table and makes ACID transactions and time travel possible
- D. An index that speeds up reads on large tables

19. What does `MERGE INTO` do?

- A. Creates a full copy of a table
- B. Performs an upsert, updating matching rows and inserting new ones in a single atomic operation
- C. Merges two schemas without writing data
- D. Combines two transaction logs into one

20. How do you query a Delta table as it was seven days ago?

- A. `SELECT * FROM t WHERE _date = date_sub(now(), 7)`
- B. `SELECT * FROM t VERSION AS OF 7`
- C. `SELECT * FROM t TIMESTAMP AS OF date_sub(current_timestamp(), 7)`
- D. `SELECT * FROM t AT (DAYS = 7)`

21. What does `VACUUM` do?

- A. Compacts small Parquet files
- B. Removes duplicate rows
- C. Permanently deletes data files that the transaction log no longer references
- D. Rebuilds the transaction log

22. What is the default retention period for `VACUUM`?

- A. 24 hours
- B. 7 days
- C. 30 days
- D. 90 days

23. How do managed and external Delta tables differ?

- A. Managed tables use Parquet; external tables use Delta
- B. For a managed table Unity Catalog controls both the metadata and the data files; for an external table only the metadata, so `DROP TABLE` leaves the files in place
- C. Only external tables support ACID transactions
- D. They behave identically

24. What does `OPTIMIZE ... ZORDER BY` achieve?

- A. Sorts the table alphabetically into one large file
- B. Compacts small files and places related values in the same files, which speeds up filters on those columns
- C. Removes duplicates and rebuilds statistics
- D. Converts a Parquet table to Delta

MODULE 05

Orchestration with Lakeflow Jobs

25. What does a repair run do?

- A. Resets every task and reruns the whole job
- B. Reruns only the failed tasks and the tasks downstream of them
- C. Sends an alert and cancels the run
- D. Rolls back the Delta table changes made by the failed run

26. Which Quartz cron expression runs a job every day at 02:30?

- A. `30 2 * * *`
- B. `2 30 * * *`
- C. `0 30 2 * * ?`
- D. `30 2 0 * * ?`

27. What is a file arrival trigger?

- A. A trigger that fires when a notebook cell raises an exception
- B. A trigger that fires when new files arrive in a specified storage location
- C. A trigger that watches the row count of a Delta table
- D. A trigger based only on Azure Event Grid notifications

28. How does a streaming table differ from a materialized view in Lakeflow Pipelines?

- A. Streaming tables can be defined only in Python; materialized views only in SQL
- B. A streaming table processes only the input data that is new since the last update; a materialized view keeps the stored result of a query up to date
- C. Only materialized views support ACID transactions
- D. There is no functional difference

29. Which feature passes parameters to a notebook when a job runs?
- A. The `%run` magic command
 - B. Widgets (`dbutils.widgets`)
 - C. Unity Catalog parameters
 - D. Cluster environment variables only
30. What is the `system.lakeflow.job_run_timeline` table used for?
- A. Storing notebook source code with its version history
 - B. Recording cluster starts and stops
 - C. Monitoring job and task runs, including durations, states and errors, with SQL
 - D. Listing active SQL warehouse connections

MODULE 06

Unity Catalog

31. What is the three-level namespace in Unity Catalog?
- A. `workspace.schema.table`
 - B. `catalog.database.file`
 - C. `catalog.schema.table`, where the last level can also be a view, volume or function
 - D. `metastore.catalog.schema`
32. How must privileges be granted to give a user access to a table?
- A. On the table only; higher levels are not needed
 - B. At the metastore level only
 - C. Top-down: `USE CATALOG`, then `USE SCHEMA`, then the privilege on the table
 - D. On the cluster, through IAM roles
33. What is a storage credential?
- A. A secret stored in a key vault
 - B. A Unity Catalog object holding a long-term cloud identity, such as an Access Connector, used to authenticate to external storage
 - C. An encrypted password for external databases
 - D. A JDBC connection string stored as a catalog property
34. Which groups can be granted privileges on Unity Catalog objects?
- A. Workspace-local groups created with `SQL CREATE GROUP`
 - B. Account groups, created in the account console or in the workspace settings under Identity and access
 - C. The workspace system group `users`
 - D. Any group that exists in the workspace

35. A participant has `ALL PRIVILEGES` on a catalog owned by the trainer, but their `GRANT SELECT ... TO` a group fails. Why?
- A. `GRANT` can be run only from a SQL warehouse
 - B. `ALL PRIVILEGES` must be granted at the metastore level before grants are allowed
 - C. `ALL PRIVILEGES` does not include `MANAGE`; only the owner, a metastore admin or a holder of `MANAGE` can grant privileges
 - D. Groups can receive privileges only on tables

MODULE 07

AI/BI dashboards and Genie

36. What is a Genie space?
- A. A GPU cluster for machine learning models
 - B. A natural-language interface in which users ask questions about data and receive answers based on generated SQL
 - C. A shared area for editing notebooks together
 - D. A dashboard for monitoring pipeline health
37. What data does a Genie space use to answer questions?
- A. Any storage path specified during setup
 - B. A Vector Search index
 - C. The Unity Catalog tables and views configured for the space
 - D. A live connection to an external PostgreSQL database
38. How can you share a published AI/BI dashboard with a colleague who is not a member of your workspace?
- A. Export it to PDF; there is no other way
 - B. Share it with them as a user of the Databricks account; they sign in with their organisational Microsoft Entra ID account
 - C. Copy the dashboard into their workspace
 - D. Dashboards cannot be shared outside the workspace
39. Which visualisations are available in AI/BI dashboards?
- A. 3D scatter plots only
 - B. Sankey diagrams only
 - C. Bar, line and pie charts, counters and tables
 - D. Tables only
40. When is Genie not the right tool?
- A. When a business user wants to explore data without writing SQL
 - B. When you need a strictly formatted report with approved KPIs
 - C. When an analyst wants to ask ad-hoc questions about sales
 - D. When you want to check a trend before a meeting

Answer key

Module	Answers
Module 01	1 A, 2 A, 3 B, 4 B, 5 B
Module 02	6 C, 7 B, 8 C, 9 C, 10 B
Module 03	11 B, 12 C, 13 B, 14 B, 15 C, 16 B
Module 04	17 B, 18 C, 19 B, 20 C, 21 C, 22 B, 23 B, 24 B
Module 05	25 B, 26 C, 27 B, 28 B, 29 B, 30 C
Module 06	31 C, 32 C, 33 B, 34 B, 35 C
Module 07	36 B, 37 C, 38 B, 39 C, 40 B

The questions reflect the platform as of September 2026. Questions about DBFS describe a legacy environment that you may still meet in existing workspaces: DBFS root and mounts are not available in accounts created after 18 December 2025, nor in any workspace provisioned after 30 September 2026. Genie spaces (questions 36 to 40) are a separate feature from Genie Code, the notebook assistant formerly called Databricks Assistant.